Lesson 1-9 • IVT, EVT, and MVT

Notes (filled in)

Objective: know when to use IVT, EVT and MVT, and justify the answer.

Table review

	IVT	EVT	MVT
Needs	continuous on $[a,b]$	continuous on a closed interval $[a,b]$	differentiable on (a,b) (so also continuous)
Proves	heights (values)	absolute max and min	slopes: $f'(c) = \frac{f(b)-f(a)}{b-a}$

Remember: differentiable $\Rightarrow$ continuous, but continuous does not imply differentiable.

Example 1

f is differentiable for all x.

x	1	3	5	8
$f(x)$	4	-2	6	10

a) Must there be a c in $(1,3)$ with $f(c) = 0$? b) Must there be a c in $(3,5)$ with $f'(c) = 4$? c) Must f have an absolute maximum on $[1,8]$?

a) Yes, IVT: f is continuous (because differentiable) and 0 is between $f(1) = 4$ and $f(3) = -2$.
b) Yes, MVT: $\frac{f(5)-f(3)}{5-3} = \frac{6-(-2)}{2} = 4$.
c) Yes, EVT: f is continuous on the closed interval $[1,8]$. The maximum is at least 10.

Example 2

$g(x) = \dfrac{1}{x-2}$ on $[1,3]$, with $g(1) = -1$ and $g(3) = 1$. Does the IVT guarantee a c with $g(c) = 0$?

No: g is not continuous on $[1,3]$ (vertical asymptote at $x = 2$), so the IVT does not apply. In fact $g(x) = 0$ has no solution.

Example 3

h is continuous on $[0,4]$, but its graph has a sharp corner at $x = 2$. Which theorems apply to h on $[0,4]$?

IVT and EVT apply (they only need continuity on $[0,4]$). The MVT does not apply, because h is not differentiable at $x = 2$.

Summary

Name the theorem, check what it needs (continuous, closed interval, or differentiable), then state what it proves: a height (IVT), an absolute max or min (EVT), or a slope (MVT).